

Ali Hassane Yaya

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AI engineer who designs and ships production-grade LLM systems — multi-agent content pipelines, model evaluation platforms, safety test harnesses, and cost-aware model routing infrastructure. One year of real startup experience building AI features used by real users; currently deepening theory through an MSc in Data Science & Business Intelligence at GISMA Berlin. Comfortable across the full stack: from system architecture and API design to frontend UIs and live monitoring dashboards. Brings a consistent bias toward observable, measurable, and cost-efficient outcomes — not just code that runs.

🎓 Education

GISMA Business School, Berlin	2025 – Present
MSc Data Science & Business Intelligence (<i>part-time</i>)	
International Islamic University Malaysia (IIUM)	Mar 2021 – Feb 2025
Bachelor of Information Technology (Hons.) — Business Intelligence & Information Science	
Relevant modules: Data Warehousing, Network Security, Software Engineering	

💻 Experience

Full-Stack Developer — XtraLesson (EdTech)	2025 – 2026
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Qatar / Remote · Online learning platform serving Arabic, French, and Swahili-speaking learners across Africa and the Middle East

- Engineered an AI translation pipeline (OpenAI GPT-4o-mini) that converted an entire English course library into three languages with chunked async processing and fault-tolerant resumption — cutting content turnaround from days to hours.

- Benchmarked and migrated the translation layer from Google Gemini to OpenAI after identifying cost overruns; added per-block ID tracking so only untranslated segments are retried, reducing redundant API calls by ~35%.
- Built a real-time job status dashboard (Started → Processing → Completed) visible to the content team, eliminating manual status-check requests and giving non-technical staff full pipeline visibility.
- Diagnosed and repaired a broken CI/CD pipeline that had been silently shipping the wrong build version for multiple releases; restored clean, predictable deployments within one sprint.

Stack: Python · FastAPI · PostgreSQL · OpenAI API · Docker · Git · CI/CD

Freelance Software Developer — Self-Employed

2023 -2024

Remote

- Built and deployed a patient triage assistant for a medical clinic — a preprocessing pipeline upstream of AI inference reduced response time by ~40%; tool adopted into daily workflow by 3 doctors.
- Developed a mental health risk prediction model on 500+ student survey responses (85% accuracy across Decision Tree, Naïve Bayes, and k-NN) with an interactive Power BI dashboard delivered to university leadership.

Stack: Python · FastAPI · React · Scikit-learn · Redis · Power BI

Projects

Multi-Agent Content Generation Pipeline ↗ [github](#)

2024

Next.js · LangGraph · OpenAI GPT-4 · Supabase · Prometheus · Grafana

- Designed a four-agent LLM pipeline — Researcher, Writer, Fact-Checker, Style-Polisher — that takes a product brief and produces a publish-ready article end-to-end with no human steps required.
- Built automatic self-correction using LangGraph: when the Fact-Checker flags an inaccuracy, the workflow re-routes to the Writer with specific feedback and retries, keeping hallucination rates low without manual review.
- Delivered a live operations dashboard (Grafana + Supabase) tracking per-run token cost, fact-check failure rate, and quality rubric scores — full pipeline observability for non-technical stakeholders.

Drift-Aware AI Retraining System

2025

Python · Supabase (pgvector) · EvidentlyAI · Prometheus · Grafana · OpenAI Fine-Tune API

- Built a monitoring service that detects model knowledge staleness by tracking shifts in user query patterns via embeddings in a vector database — catching drift before users notice quality degradation.
- Automated recovery: detected drift triggers re-indexing of the knowledge base or launches an OpenAI fine-tuning job automatically — the system heals itself without requiring engineer intervention.
- Live Grafana dashboard shows model accuracy and API cost before and after each retraining event, with Prometheus alerts when any drift metric passes its threshold.

LLM Model Router with Budget Controls

2025

Next.js · OpenAI API · Anthropic API · Redis (Upstash) · Supabase · Grafana

- Built an API gateway that classifies each query as easy, medium, or hard and routes it to the cheapest capable model — directing 80% of traffic to lower-cost tiers with no measurable quality drop.
- Enforced a configurable monthly spend cap via atomic Redis counters; at 90% usage, expensive model routes shut off automatically and the team receives a Slack alert — eliminating surprise invoices.
- Every routing decision, token count, and cost logged to Supabase and visualised in a Grafana cost-vs-quality heatmap, turning AI spend from guesswork into tunable, data-backed decisions.

Guardrail & Red-Team Test Harness

2026

Python · OpenAI Moderation API · Supabase · Next.js · GitHub Actions

- Built a 50+ adversarial prompt suite covering jailbreaks, prompt injections, and PII-extraction attempts; each runs through a three-layer defence: system-prompt rules → OpenAI Moderation API → regex PII scrubbers.
- Automated nightly regression tests via GitHub Actions — each run computes jailbreak success rate, false positives, and false negatives, diffing against the prior run so new vulnerabilities are caught before production.
- Reduced jailbreak success rate from 18% to under 4% across three improvement cycles; every failure logged in a live incident dashboard with full prompt detail for post-mortem review.

Evaluation-as-a-Service (EaaS) Platform

2026

Next.js · Python · Supabase · OpenAI GPT-4-as-Judge · Vercel · GitHub Actions

- Built an internal platform where teams upload a CSV of model outputs, choose a grading method (exact-match, BLEU, or GPT-4-as-Judge), and receive per-item scores with explanations in minutes — replacing a manual QA process that took days.

- Integrated a CI/CD deployment gate: if a new model version scores below a configured threshold, the deployment is blocked automatically — preventing quality regressions from reaching end users.
- Side-by-side version comparison and multi-run trend charts give teams clear, data-backed evidence of whether prompt changes or new models genuinely improve output quality over time.

Skills

Programming: Python, JavaScript / TypeScript, SQL, PHP, R

AI & LLMs: OpenAI API, Anthropic API, LangChain, LangGraph, CrewAI, pgvector, EvidentlyAI, Prompt Engineering

MLOps: Prometheus, Grafana, Embedding Drift Detection, LLM Evaluation, Model Routing, Adversarial Red-Teaming

Backend: FastAPI, Node.js, Express.js, Next.js API Routes, REST APIs

Frontend: React, Next.js, HTML5, CSS3, Tailwind CSS

Databases: PostgreSQL, Supabase, MySQL, MongoDB, Redis (Upstash)

DevOps & Cloud: Docker, Git, GitHub Actions, CI/CD, AWS (Cloud Practitioner level), Vercel

BI & Analytics: Power BI, Tableau, Scikit-learn, Pandas, NumPy

Languages: Arabic (Native), English (Fluent), French (Intermediate), Bahasa Melayu (Basic)

Certificates

- AWS Cloud Practitioner (*in progress — exam scheduled*)
- REST APIs & FastAPI — Udemy
- Introduction to Cybersecurity — Cisco Networking Academy

Honors, Awards & Activities

- **1st Place** — CTF Cybersecurity Competition, IIUM (May 2022)
- Top 20% — PicoCTF Global Leaderboard (2023)
- Solved 15+ HackTheBox beginner machines (2024–2025)
- Volunteer Translator (Arabic → French) — Islamic Development Bank (2023)